A Technical Introduction to CycleGAN

Section 1

Original GAN, non-saturating loss: This objective for generator was recommended in the original paper for faster convergence. L G = E x ∼ μ G ⁡ [ln ⁡ D (x)] {\displaystyle L_{G}=\operatorname {E} _{x\sim \mu _{G}}[\ln D(x)]} The effect of using this objective is analyzed in Section 2.2.2 of Arjovsky et al. Original GAN, maximum likelihood:

Section 2

BigGAN The BigGAN is essentially a self-attention GAN trained on a large scale (up to 80 million parameters) to generate large images of ImageNet (up to 512 x 512 resolution), with numerous engineering tricks to make it converge.

Section 3

In practice The generative network generates candidates while the discriminative network evaluates them. This creates a contest based on data distributions, where the generator learns to map from a latent space to the true data distribution, aiming to produce candidates that the discriminator cannot distinguish from real data. The discriminator's goal is to correctly identify these candidates, but as the generator improves, its task becomes more challenging, increasing the discriminator's error rate. A known dataset serves as the initial training data for the discriminator. Training involves presenting it with samples from the training dataset until it achieves acceptable accuracy. The generator is trained based on whether it succeeds in fooling the discriminator. Typically, the generator is seeded with randomized input that is sampled from a predefined latent space (e.g. a multivariate normal distribution). Thereafter, candidates synthesized by the generator are evaluated by the discriminator. Independent backpropagation procedures are applied to both networks so that the generator produces better samples, while the discriminator becomes more skilled at flagging synthetic samples. When used for image generation, the generator is typically a deconvolutional neural network, and the discriminator is a convolutional neural network.

Section 4

StyleGAN-2 StyleGAN-2 improves upon StyleGAN-1, by using the style latent vector to transform the convolution layer's weights instead, thus solving the "blob" problem. This was updated by the StyleGAN-2-ADA ("ADA" stands for "adaptive"), which uses invertible data augmentation as described above. It also tunes the amount of data augmentation applied by starting at zero, and gradually increasing it until an "overfitting heuristic" reaches a target level, thus the name "adaptive".

Section 5

The discriminator's strategy set is the set of Markov kernels μ D : (Ω , B) → P ([0 , 1] , B ([0 , 1])) {\displaystyle \mu _{D}:(\Omega ,{\mathcal {B}})\to {\mathcal {P}}([0,1],{\mathcal {B}}([0,1]))} , where B ([0 , 1]) {\displaystyle {\mathcal {B}}([0,1])} is the Borel σ-algebra on [0 , 1] {\displaystyle [0,1]} .Since issues of measurability never arise in practice, these will not concern us further.

Section 6

Main theorems for GAN game The original GAN paper proved the following two theorems: Interpretation: For any fixed generator strategy μ G {\displaystyle \mu _{G}} , the optimal discriminator keeps track of the likelihood ratio between the reference distribution and the generator distribution: D (x) 1 − D (x) = d μ ref d μ G (x) = μ ref (d x) μ G (d x) ; D (x) = σ (ln ⁡ μ ref (d x) − ln ⁡ μ G (d x)) {\displaystyle {\frac {D(x)}{1-D(x)}}={\frac {d\mu _{\text{ref}}}{d\mu _{G}}}(x)={\frac {\mu _{\text{ref}}(dx)}{\mu _{G}(dx)}};\quad D(x)=\sigma (\ln \mu _{\text{ref}}(dx)-\ln \mu _{G}(dx))} where σ {\displaystyle \sigma } is the logistic function. In particular, if the prior probability for an image x {\displaystyle x} to come from the reference distribution is equal to 1 2 {\displaystyle {\frac {1}{2}}} , then D (x) {\displaystyle D(x)} is just the posterior probability that x {\displaystyle x} came from the reference distribution: D (x) = Pr (x came from reference distribution ∣ x) . {\displaystyle D(x)=\Pr(x{\text{ came from reference distribution}}\mid x).}

Section 7

Generator-encoder team aims to minimize the objective, and discriminator aims to maximize it: min G , E max D L (G , E , D) {\displaystyle \min _{G,E}\max _{D}L(G,E,D)} In the paper, they gave a more abstract definition of the objective as: L (G , E , D) = E (x , z) ∼ μ E , X [ln ⁡ D (x , z)] + E (x , z) ∼ μ G , Z [ln ⁡ (1 − D (x , z))] {\displaystyle L(G,E,D)=\mathbb {E} _{(x,z)\sim \mu _{E,X}}[\ln D(x,z)]+\mathbb {E} _{(x,z)\sim \mu _{G,Z}}[\ln(1-D(x,z))]} where μ E , X (d x , d z) = μ X (d x) ⋅ δ E (x) (d z) {\displaystyle \mu _{E,X}(dx,dz)=\mu _{X}(dx)\cdot \delta _{E(x)}(dz)} is the probability distribution on Ω X × Ω Z {\displaystyle \Omega _{X}\times \Omega _{Z}} obtained by pushing μ X {\displaystyle \mu _{X}} forward via x ↦ (x , E (x)) {\displaystyle x\mapsto (x,E(x))} , and μ G , Z (d x , d z) = δ G (z) (d x) ⋅ μ Z (d z) {\displaystyle \mu _{G,Z}(dx,dz)=\delta _{G(z)}(dx)\cdot \mu _{Z}(dz)} is the probability distribution on Ω X × Ω Z {\displaystyle \Omega _{X}\times \Omega _{Z}} obtained by pushing μ Z {\displaystyle \mu _{Z}} forward via z ↦ (G (x) , z) {\displaystyle z\mapsto (G(x),z)} . Applications of bidirectional models include semi-supervised learning, interpretable machine learning, and neural machine translation.

Section 8

L G = E x ∼ μ G ■ [(D (x) − c) 2] {\displaystyle L_{G}=\operatorname {E} _{x\sim \mu _{G}}[(D(x)-c)^{2}]} where a , b , c {\displaystyle a,b,c} are parameters to be chosen. The authors recommended a = − 1 , b = 1 , c = 0 {\displaystyle a=-1,b=1,c=0} .

Section 9

Concerns have been raised about the potential use of GAN-based human image synthesis for sinister purposes, e.g., to produce fake, possibly incriminating, photographs and videos. GANs can be used to generate unique, realistic profile photos of people who do not exist, in order to automate creation of fake social media profiles. In 2019 the state of California considered and passed on October 3, 2019, the bill AB-602, which bans the use of human image synthesis technologies to make fake pornography without the consent of the people depicted, and bill AB-730, which prohibits distribution of manipulated videos of a political candidate within 60 days of an election. Both bills were authored by Assembly member Marc Berman and signed by Governor Gavin Newsom. The laws went into effect in 2020. DARPA's Media Forensics program studies ways to counteract fake media, including fake media produced using GANs.